

VECTOR SPACE

Dr. Juhi Kesarwani

Dr. Ashish Kumar Kesarwany

Mathematics Division, School of Advanced Sciences and Languages
VIT Bhopal University, Bhopal-Indore Highway, Kothrikalan, Sehore, Madhya Pradesh, 466114, India

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Vector Space

For $u, v, w, \in \mathbb{R}^n$ and $c, d \in \mathbb{R}$, we have

$$\textcircled{1} \quad u + v \in \mathbb{R}^n;$$

$$\textcircled{2} \quad u + v = v + u;$$

$$\textcircled{3} \quad (u + v) + w = u + (v + w);$$

$$\textcircled{4} \quad u + 0 = u;$$

$$\textcircled{5} \quad u + (-u) = 0;$$

Vector Space

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- ② $u + v = v + u$;
- ③ $(u + v) + w = u + (v + w)$;
- ④ $u + 0 = u$;
- ⑤ $u + (-u) = 0$;
- ⑥ $c \cdot u \in \mathbb{R}^n$;
- ⑦ $c \cdot (u + v) = c \cdot u + c \cdot v$;
- ⑧ $(c + d) \cdot u = c \cdot u + d \cdot u$;
- ⑨ $c \cdot (d \cdot u) = (cd) \cdot u$; and
- ⑩ $1 \cdot u = u$.

The above properties are sufficient to do vector algebra in $\mathbb{R}^n$.



If $u, v, w, 0 \in \mathbb{C}^n$ and $c, d \in \mathbb{R}$, we get all the previous ten properties.



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If $A, B, C, O \in M_2(\mathbb{R})$ (set of all 2×2 real matrices) and $c, d \in \mathbb{R}$, we get all the previous ten properties.



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If $A, B, C, O \in M_2(\mathbb{R})$ (set of all 2×2 real matrices) and $c, d \in \mathbb{R}$, we get all the previous ten properties.



If $p(x), q(x), r(x), 0 \in \mathbb{R}_2[x]$ (set of all polynomials of degree at most two with real coefficients) and $c, d \in \mathbb{R}$, we get all the previous ten properties.

Note

The symbol $\mathbb{F}$ will be used as a representative of $\mathbb{R}$ or $\mathbb{C}$. That is, $\mathbb{F} = \mathbb{R}$ or $\mathbb{F} = \mathbb{C}$.

Definition

Let $V(\neq \phi)$ be a set. For every $u, v \in V$ and $c \in \mathbb{F}$, let

✿ $u + v$ (called the vector addition) be defined; and

✿ $c \cdot u$ (called the scalar multiplication) be defined.

Then V is called a vector space over $\mathbb{F}$ **if** for all $u, v, w \in V$ and for all $c, d \in \mathbb{F}$, the following properties are satisfied:

Definition cont. . .

- ① $u + v \in V$;
- ② $u + v = v + u$;
- ③ $(u + v) + w = u + (v + w)$;
- ④ There is an element 0 , called a zero element, such that $u + 0 = u$;
- ⑤ For each $u \in V$, there is an element $-u$, called the additive inverse of u , such that $u + (-u) = 0$;

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- ⑤ For each $u \in V$, there is an element $-u$, called the additive inverse of u , such that $u + (-u) = 0$;
- ⑥ $c \cdot u \in V$;
- ⑦ $c \cdot (u + v) = c \cdot u + c \cdot v$;
- ⑧ $(c + d) \cdot u = c \cdot u + d \cdot u$;
- ⑨ $c \cdot (d \cdot u) = (cd) \cdot u$; and
- ⑩ $1 \cdot u = u$.

Note

✿ The elements of V are called **vectors**.

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Example

For any $n \geq 1$, the set $\mathbb{R}^n$ is a vector space over $\mathbb{R}$ with respect to usual operations of addition and scalar multiplication.

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The set $\mathbb{Z}$ is not a vector space over $\mathbb{R}$ with respect to usual operations of addition and scalar multiplication.

Example

Let $\mathbb{R}_2[x] = \{a + bx + cx^2 : a, b, c \in \mathbb{R}\}$. For $p(x) = a_0 + b_0x + c_0x^2$, $q(x) = a_1 + b_1x + c_1x^2$ and $k \in \mathbb{R}$, define

$$p(x) + q(x) = (a_0 + a_1) + (b_0 + b_1)x + (c_0 + c_1)x^2$$

$$k \cdot p(x) = (ka_0) + (kb_0)x + (kc_0)x^2.$$

Then $\mathbb{R}_2[x]$ is a vector space over $\mathbb{R}$.

Example

The set $M_2(\mathbb{R})$ of all 2×2 real matrices is a vector space over $\mathbb{R}$ with respect to usual operations of matrix addition and matrix scalar multiplication.

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Example

The set $\mathbb{R}^2$ is not a vector space over $\mathbb{R}$ with respect to usual operations of addition and the following definition of scalar multiplication:

$$c \cdot \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} cx \\ 0 \end{bmatrix} \quad \text{for } \begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2, c \in \mathbb{R}.$$

Note

- ✿ If there is no confusion, $c \cdot u$ is simply written as cu .
- ✿ We write $V(\mathbb{F})$ to denote that V is a vector space over $\mathbb{F}$.
- ✿ If $\mathbb{F}$ needs to be specified, then we write $V(\mathbb{F})$ to denote that V is a vector space over $\mathbb{F}$.
- ✿ We call V a real vector space or complex vector space according as $\mathbb{F} = \mathbb{R}$ or $\mathbb{F} = \mathbb{C}$.

Result

Let V be a vector space over $\mathbb{F}$. Let $u \in V$ and $c \in \mathbb{F}$. Then

- ① $0 \cdot u = 0$;
- ② $c \cdot 0 = 0$;
- ③ $(-1) \cdot u = -u$; and
- ④ If $c \cdot u = 0$ then either $c = 0$ or $u = 0$.

Example

Let $\mathbb{R}_+$ denotes the set of positive real numbers. Determine whether $\mathbb{R}_+$ forms a vector space over $\mathbb{R}$ under the addition $\oplus$ and scalar multiplication $\odot$, where $x \oplus y = xy$ and $c \odot x = x^c$.

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① For $x, y \in \mathbb{R}_+$, we have

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which gives $x \oplus y = y \oplus x$ for all $x, y \in \mathbb{R}_+$.

③ For $x, y, z \in \mathbb{R}_+$, consider

$$x \oplus (y \oplus z) = x \oplus (yz) = x(yz) = (xy)z = x \oplus (yz) = x \oplus (y \oplus z).$$

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④ Suppose α is the zero element, then by definition, for all $x \in \mathbb{R}_+$, we have

$$x \oplus \alpha = x \implies \alpha x = x \implies \alpha = 1.$$

Thus, 1 is the zero element of $\mathbb{R}_+$.

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$$x \oplus y = 1 \implies xy = 1 \implies y = \frac{1}{x}.$$

Thus, $\frac{1}{x}$ is the additive inverse of x .

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⑥ For $c \in \mathbb{R}$ and $x \in \mathbb{R}_+$, we have

$$c \odot x = x^c \in \mathbb{R}_+.$$

Thus, $c \odot x \in \mathbb{R}_+$ for all $x \in \mathbb{R}_+$ and for all $c \in \mathbb{R}$.

⑦ For $c \in \mathbb{R}$ and $x, y \in \mathbb{R}_+$, consider

$$c \odot (x \oplus y) = c \odot (xy) = (xy)^c = x^c y^c = x^c \oplus y^c = (c \odot x) \oplus (c \odot y).$$

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⑧ For $c, d \in \mathbb{R}$ and $x \in \mathbb{R}_+$, consider

$$(c + d) \odot x = x^{(c+d)} = x^c x^d = x^c \oplus x^d = (c \odot x) \oplus (d \odot x).$$

Thus, $(c + d) \odot x = (c \odot x) \oplus (d \odot x)$, for all $c, d \in \mathbb{R}$ and for all $x \in \mathbb{R}_+$.

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⑨ For $c, d \in \mathbb{R}$ and $x \in \mathbb{R}_+$, consider

$$c \odot (d \odot x) = c \odot x^d = (x^d)^c = x^c d = (cd) \odot x.$$

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Thus, $c \odot (d \odot x) = (cd) \odot x$, for all $c, d \in \mathbb{R}$ and for all $x \in \mathbb{R}_+$.

⑩ $1 \odot x = x^1 = x$ for all $x \in \mathbb{R}_+$.

⑦ For $c \in \mathbb{R}$ and $x, y \in \mathbb{R}_+$, consider

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⑨ For $c, d \in \mathbb{R}$ and $x \in \mathbb{R}_+$, consider

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Thus, $c \odot (d \odot x) = (cd) \odot x$, for all $c, d \in \mathbb{R}$ and for all $x \in \mathbb{R}_+$.

⑩ $1 \odot x = x^1 = x$ for all $x \in \mathbb{R}_+$.

Above ten points shows that $\mathbb{R}_+(\mathbb{R})$ is a vector space with given addition and scalar multiplication.

Exercise 1:

Let $V = \mathbb{R}^2$ and define addition $\oplus$ and scalar multiplication $\odot$ in V as

$$\begin{bmatrix} a \\ b \end{bmatrix} \oplus \begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} a + 2c \\ b + 2d \end{bmatrix} \quad \text{and} \quad k \odot \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} ka \\ kb \end{bmatrix}$$

Determine whether V is a vector space. Also find additive identity and additive inverse.

Exercise 2:

Suppose that V is a vector space. Show that

- (i) A zero vector of V is unique
- (ii) A negative of a vector v in V is unique.

Exercise 3:

Let V be the set of all ordered pairs (x, y) , where x, y are real number. Let $a = (x_1, y_1)$ and $b = (x_2, y_2)$ be elements in V . Define the addition as

$$(x_1, y_1) + (x_2, y_2) = (x_1 x_2, y_1 y_2)$$

and the scalar multiplication as

$$\alpha(x_1, y_1) = (\alpha x_1, \alpha y_1).$$

Show that V is not a vector space.

Exercise 4:

Let $V = \mathbb{R}^2$, with addition and scalar multiplication defined by

$$\begin{bmatrix} x \\ y \end{bmatrix} \oplus \begin{bmatrix} w \\ z \end{bmatrix} = \begin{bmatrix} x + w \\ y + z \end{bmatrix} \quad \text{and} \quad a \odot \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} ax + a \\ ay - 2a \end{bmatrix}.$$

Determine whether V is a vector space or not? If yes, find additive identity and inverse.

Exercise 5:

Determine whether the given set is a vector space. If not, give at least one axiom that is not satisfied. Unless otherwise stated, assume that vector addition and scalar multiplication are the ordinary operations defined on the set.

(a) The set of vectors $\left\{ \begin{bmatrix} a \\ b \end{bmatrix} \in \mathbb{R}^2 : b = 3a + 1 \right\}$.

(b) $\mathbb{R}^2$ with scalar multiplication defined by $k \cdot \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} ka \\ b \end{bmatrix}$.

(c) $\mathbb{R}^2$ with scalar multiplication defined by $k \cdot \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} ka \\ 0 \end{bmatrix}$.

(d) The set of real numbers, with addition defined by $x + y = x - y$.

(e) The set $\mathbb{R}^3$, with the vector addition operation $\oplus$ and scalar multiplication $\odot$ defined by

$$\begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} \oplus \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} a_1 + b_1 + 5 \\ a_2 + b_2 - 7 \\ a_3 + b_3 + 1 \end{bmatrix} \quad \text{and} \quad k \odot \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} ka_1 + 5(k-1) \\ ka_2 - 7(k-1) \\ ka_3 + k - 1 \end{bmatrix}$$

Subspace

Definition(Subspace)

Let V be a vector space over $\mathbb{F}$ and $(\phi \neq) W \subseteq V$. Then W is called a **subspace** of V if and only if $au + bv \in W$ for every $u, v \in W$ and for every $a, b \in \mathbb{F}$.

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Remark

- ✿ If W is a subspace of a vector space V , then W is also a vector space.
- ✿ If W is a subspace of a vector space V then $0 \in W$.
- ✿ The sets $\{0\}$ and V are always subspaces of any vector space V . These are called the **trivial subspaces**.

Example

Let W be the set of all 2×2 real symmetric matrices. Then W is a subspace of $M_2(\mathbb{R})$.

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Solution

Since $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}^T = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ and hence $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \in W$. This gives $W \neq \phi$. Let $A, B \in W$ and $a, b \in \mathbb{R}$ then

$$A^T = A \quad \text{and} \quad B^T = B$$

Consider

$$(aA + bB)^T = (aA)^T + (bB)^T = aA^T + bB^T = aA + bB.$$

Thus, $aA + bB \in W$ and hence it shows that W is a subspace.

Example

Let $W = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3 : x + y - z = 0 \right\}$. Then W is a subspace of $\mathbb{R}^3$.

Result

Let V be a vector space.

- ✿ If U and W are subspaces of V , then $U \cap W$ is also a subspace of V .
- ✿ If $\{U_i\}$ is a collection of subspaces of V , then $\cap_i U_i$ is also a subspace of V .
- ✿ Let U and W be subspaces of V . Then $U \cup W$ is a subspace of V iff $U \subseteq W$ or $W \subseteq U$.

Exercise 6:

Each of the following sets are not a subspace of the specified vector space. For each set, give a reason why it is not a subspace.

$$(a) S_1 = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \mathbb{R}^3 \mid x_1 \geq 0 \right\} \text{ in the vector space } \mathbb{R}^3.$$

$$(b) S_2 = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \mathbb{R}^3 \mid x_1 - 4x_2 + 5x_3 = 2 \right\} \text{ in the vector space } \mathbb{R}^3.$$

$$(c) S_3 = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2 \mid y = x^2 \right\} \text{ in the vector space } \mathbb{R}^2.$$

(d) Let $\mathbb{R}_4[x]$ be the vector space of all polynomials of degree 4 or less with real coefficients.

$$S_4 = \{f(x) \in \mathbb{R}_4[x] \mid f(1) \text{ is an integer}\}$$

in the vector space $\mathbb{R}_4[x]$.

(e) $S_5 = \{f(x) \in \mathbb{R}_4[x] \mid f(1) \text{ is a rational number}\}$ in the vector space $\mathbb{R}_4[x]$.

(f) Let $M_{2 \times 2}$ be the vector space of all 2×2 real matrices.

$$S_6 = \{A \in M_{2 \times 2} \mid \det(A) \neq 0\}$$

in the vector space $M_{2 \times 2}$.

(g) $S_7 = \{A \in M_{2 \times 2} \mid \det(A) = 0\}$ in the vector space $M_{2 \times 2}$.

Exercise 7:

Determine whether the following sets form subspaces of $\mathbb{R}^2$:

$$(a) \left\{ \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} : x_1 + x_2 = 0 \right\}$$

$$(d) \left\{ \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} : |x_1| = |x_2| \right\}$$

$$(b) \left\{ \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} : x_1 x_2 = 0 \right\}$$

$$(e) \left\{ \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} : x_1^2 = x_2^2 \right\}$$

$$(c) \left\{ \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} : x_1 = 3x_2 \right\}$$

Exercise 8:

Determine whether the following sets form subspaces of $\mathbb{R}^3$:

$$(a) \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} : x_1 + x_3 = 1 \right\}$$

$$(c) \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} : x_3 = x_1 + x_2 \right\}$$

$$(b) \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} : x_1 = x_2 = x_3 \right\}$$

$$(d) \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} : x_3 = x_1 \text{ or } x_3 = x_2 \right\}$$

Exercise 9:

Determine whether the following are subspaces of $M_2(\mathbb{R})$

- (a) The set of all 2×2 diagonal matrices
- (b) The set of all 2×2 triangular matrices
- (c) The set of all 2×2 lower triangular matrices
- (d) The set of all 2×2 matrices $A = [a_{ij}]$ such that $a_{12} = 1$.
- (e) The set of all 2×2 matrices $B = [b_{ij}]$ such that $b_{11} = 0$.
- (f) The set of all symmetric 2×2 matrices
- (g) The set of all singular 2×2 matrices

Exercise 10:

In each part, V is a vector space and S is a subset of V . Determine whether S is a subspace of V .

- (a) $V = \mathbb{R}^3$ and $S = \left\{ \begin{bmatrix} x \\ 12 \\ 3x \end{bmatrix} \mid x \in \mathbb{R} \right\}$
- (b) $V = \mathbb{R}^2$ and $S = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} \mid 2x - 5y = 11 \right\}$
- (c) $V = \mathbb{R}^n$ and $S = \{ \vec{x} \in \mathbb{R}^n \mid A\vec{x} = 2\vec{x} \}$, where A is a particular $n \times n$ matrix.

Exercise 11:

Determine whether the following are subspaces of $\mathbb{R}_4[x]$:

- (a) The set of polynomials in $\mathbb{R}_4[x]$ of even degree.
- (b) The set of all polynomials of degree 3.
- (c) The set of all polynomials $p(x)$ in $\mathbb{R}_4[x]$ such that $p(0) = 0$.
- (d) The set of all polynomials in $\mathbb{R}_4[x]$ having at least one real root

Exercise 12:

Let V be the vector space over $\mathbb{R}$ consisting of all $n \times n$ real matrices for some fixed integer n . Prove or disprove that the following subsets of V are subspaces of V .

- (a) The set S consisting of all $n \times n$ symmetric matrices.
- (b) The set T consisting of all $n \times n$ skew-symmetric matrices.
- (c) The set U consisting of all $n \times n$ non-singular matrices.

Exercise 13:

Let M_n denotes the vector space of all $n \times n$ real matrices. Among the following subsets of M_n , decide which are subspace of M_n . If a subset is subspace, prove it. If not, justify your argument with a counterexample.

- (a) $S_1 = \{A \in M_n : A \text{ is non-singular} \}$
- (b) $S_2 = \{A \in M_n : \det(A) = 0\}$
- (c) $S_3 = \{A \in M_n : \text{trace}(A) = 0\}$
- (d) $S_4 = \{BA : A \in M_n\}$, where B is some fixed matrix in M_n .

Linear Combination

Let V be a vector space over $\mathbb{F}$. Let $v_1, v_2, \dots, v_k$ be elements in V . A **linear combination** of these elements is any expression of the form

$$\alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_k v_k$$

where the coefficients $\alpha_1, \alpha_2, \dots, \alpha_k$ are scalars.

Example

Consider the vector space $\mathbb{R}^3$ over $\mathbb{R}$. Let $v_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, $v_2 = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$. Express $u = \begin{bmatrix} -1 \\ 2 \\ -1 \end{bmatrix}$ as a linear combination of v_1 and v_2 .

Example

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Solution

We must find scalars a_1 and a_2 such that $u = a_1 v_1 + a_2 v_2$. Thus

$$a_1 + a_2 = -1$$

$$2a_1 = 2$$

$$3a_1 + 2a_2 = -1$$

solving these equations, we get $a_1 = 1$ and $a_2 = -2$. Thus

$$u = v_1 - 2v_2$$

Example

Consider the vector space $\mathbb{R}^3$ over $\mathbb{R}$. Let $v_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, $v_2 = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$. Express $v = \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix}$ as a linear combination of v_1 and v_2 .

Example

Consider the vector space $\mathbb{R}^3$ over $\mathbb{R}$. Let $v_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, $v_2 = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$. Express $v = \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix}$ as a linear combination of v_1 and v_2 .

Solution

We must find scalars b_1 and b_2 such that $v = b_1 v_1 + b_2 v_2$. Thus

$$b_1 + b_2 = -1$$

$$2b_1 = 2$$

$$3b_1 + 2b_2 = 0$$

The above system of linear equations has no solution, so v can not be expressed as a linear combination of v_1 and v_2 .

Exercise 14:

Express the polynomial $p(x) = x^2 + 4x - 3$ in $\mathbb{R}_3[x]$ as a linear combination of the polynomials $p_1(x) = x^2 - 2x + 5$, $p_2(x) = 2x^2 - 3x$, $p_3(x) = x + 1$.

Spanning set

✿ Let V be a vector space over $\mathbb{F}$. Let $S = \{v_1, v_2, \dots, v_k\}$ be a subset of V . Then the set of all linear combinations of $v_1, v_2, \dots, v_k$ is called the **span** of $v_1, v_2, \dots, v_k$, and is denoted by **$\text{span}(v_1, v_2, \dots, v_k)$** or **$\text{span}(S)$** . That is,

$$\text{span}(S) = \{c_1 v_1 + c_2 v_2 + \dots + c_k v_k \mid c_1, c_2, \dots, c_k \in \mathbb{F}\}.$$

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- ✿ Let $S \subseteq V$ (**may be infinite!**). The **span of S** is defined by

$$\text{span}(S) := \left\{ \sum_{i=1}^m \alpha_i v_i \mid v_i \in S, \alpha_i \in \mathbb{F}, m \text{ a non-negative integer} \right\}.$$

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Remark

Convention: **$\text{span}(\phi) = \{0\}$**

Result

Let S be a subset of a vector space V . Then $\text{span}(S)$ is a subspace of V .

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$$\mathbb{R}_2[x] = \text{span}(1, x, x^2).$$

Result

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Example

Consider the vector space $\mathbb{R}_2[x]$. Then

$$\mathbb{R}_2[x] = \text{span}(1, x, x^2).$$

Example

Consider $\mathbb{R}[x]$ = set of all polynomials in x over $\mathbb{R}$. Then

$$\mathbb{R}[x] = \text{span}(1, x, x^2, \dots)$$

Example

Find $\text{span}(v_1, v_2)$, where $v_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, $v_2 = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$.

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Solution

$\text{span}(v_1, v_2)$ is the set of all vectors $\begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3$ such that $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = a_1 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + a_2 \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$.

We wish to know for what values of $\begin{bmatrix} x \\ y \\ z \end{bmatrix}$ does this system of equations have solutions for a_1 and a_2 . Thus

$$a_1 + a_2 = x$$

$$2a_1 = y$$

$$3a_1 + 2a_2 = z$$

Solution cont. . .

The augmented matrix of above system of linear equations is

$$\left[\begin{array}{cc|c} 1 & 1 & x \\ 2 & 0 & y \\ 3 & 2 & z \end{array} \right] \begin{array}{l} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - 3R_1 \end{array} \quad \left[\begin{array}{cc|c} 1 & 1 & x \\ 0 & -2 & y - 2x \\ 0 & -1 & z - 3x \end{array} \right] R_3 \rightarrow R_3 - \frac{R_2}{2}$$
$$\left[\begin{array}{cc|c} 1 & 1 & x \\ 0 & -2 & y - 2x \\ 0 & 0 & z - 2x - \frac{y}{2} \end{array} \right]$$

The above system of linear equations has solution if $z - 2x - \frac{y}{2} = 0$, i.e., $4x + y - 2z = 0$. Thus $\text{span}(v_1, v_2)$ is the plane $4x + y - 2z = 0$.

$$\text{span}(v_1, v_2) = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3 : 4x + y - 2z = 0 \right\}$$

Linear Dependence and Linear Independence

Linear Dependence

A set $\{v_1, v_2, \dots, v_k\}$ of vectors in a vector space V is said to be **linearly dependent (LD)** if there are scalars $\alpha_1, \alpha_2, \dots, \alpha_k$, **at least one of them non-zero**, such that

$$\alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_k v_k = 0.$$

An **infinite set** $S \subseteq V$ is **linearly dependent** if there is some finite linearly dependent subset of S .

Linear Independence

The set S of vectors in a vector space V is said to be **linearly independent (LI)** if it is **not** linearly dependent. Thus if $S = \{v_1, v_2, \dots, v_k\}$ then S is **linearly independent (LI)** if

$$\alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_k v_k = 0 \implies \alpha_1 = \alpha_2 = \dots = \alpha_k = 0;$$

If S is **infinite** then S is **linearly independent (LI)** if every **finite subset** of S is linearly independent.

Linear Independence

The set S of vectors in a vector space V is said to be **linearly independent (LI)** if it is **not** linearly dependent. Thus if $S = \{v_1, v_2, \dots, v_k\}$ then S is **linearly independent (LI)** if

$$\alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_k v_k = 0 \implies \alpha_1 = \alpha_2 = \dots = \alpha_k = 0;$$

If S is **infinite** then S is **linearly independent (LI)** if every **finite subset** of S is linearly independent.

Remark

- ✿ Set $\{0\}$ is linearly dependent as $1 \cdot 0 = 0$. [A non-trivial linear combination of 0 is 0.]
- ✿ If $0 \in S$, then S is always linearly dependent as S contains a LD set $\{0\}$.

Example

The set $\{A, B, C\}$, where $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}$, $C = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}$, is linearly dependent as

$$1 \cdot A + 1 \cdot B + (-2) \cdot C = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Example

The set $\{A, B, C\}$, where $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}$, $C = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}$, is linearly dependent as

$$1 \cdot A + 1 \cdot B + (-2) \cdot C = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Example

The set $\{1, x, x^2, \dots, x^n\}$ is linearly independent in $\mathbb{R}_n[x]$ as

$$\alpha_0 \cdot 1 + \alpha_1 \cdot x + \dots + \alpha_n \cdot x^n = 0 \implies \alpha_0 = \alpha_1 = \dots = \alpha_n = 0$$

Example

The set $\{A, B, C\}$, where $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}$, $C = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}$, is linearly dependent as

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Example

The set $\{1, x, x^2, \dots, x^n\}$ is linearly independent in $\mathbb{R}_n[x]$ as

$$\alpha_0 \cdot 1 + \alpha_1 \cdot x + \dots + \alpha_n \cdot x^n = 0 \implies \alpha_0 = \alpha_1 = \dots = \alpha_n = 0$$

Example

The set $\{1, x, x^2, \dots\}$ is linearly independent in $\mathbb{R}[x]$.

Result

- ✿ A superset of Linearly dependent set is Linearly dependent.
- ✿ A subset of Linearly independent set is Linearly independent.

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How to check whether a set of vectors of a vector space is LI or LD?

Let $S = \{v_1, v_2, \dots, v_k\}$ be a set of vectors in a vector space V . Then to check whether S is LI or LD, we need to follow the following steps:

Step-1 Start with $\alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_k v_k = 0$. This gives a homogeneous system of linear equations in variables $\alpha_1, \alpha_2, \dots, \alpha_k$.

Step-2 If above homogeneous system of linear equations has only **trivial** solution then S is LI otherwise LD.

Example

Are the vectors $v_1 = \begin{bmatrix} 1 \\ 4 \\ 0 \end{bmatrix}$, $v_2 = \begin{bmatrix} 10 \\ 2 \\ 1 \end{bmatrix}$, $v_3 = \begin{bmatrix} -5 \\ 0 \\ 6 \end{bmatrix}$ linearly dependent or independent?

Example

Are the vectors $v_1 = \begin{bmatrix} 1 \\ 4 \\ 0 \end{bmatrix}$, $v_2 = \begin{bmatrix} 10 \\ 2 \\ 1 \end{bmatrix}$, $v_3 = \begin{bmatrix} -5 \\ 0 \\ 6 \end{bmatrix}$ linearly dependent or independent?

Solution

Consider

$$\alpha_1 \begin{bmatrix} 1 \\ 4 \\ 0 \end{bmatrix} + \alpha_2 \begin{bmatrix} 10 \\ 2 \\ 1 \end{bmatrix} + \alpha_3 \begin{bmatrix} -5 \\ 0 \\ 6 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

This gives the following homogeneous system of linear equations

$$\alpha_1 + 10\alpha_2 - 5\alpha_3 = 0$$

$$4\alpha_1 + 2\alpha_2 = 0$$

$$\alpha_2 + 6\alpha_3 = 0$$

Solution cont. . .

The augmented matrix of above homogeneous system of linear equations is

$$\left[\begin{array}{ccc|c} 1 & 10 & -5 & 0 \\ 4 & 2 & 0 & 0 \\ 0 & 1 & 6 & 0 \end{array} \right] R_2 \rightarrow R_2 - 4R_1 \quad \left[\begin{array}{ccc|c} 1 & 10 & -5 & 0 \\ 0 & -38 & 20 & 0 \\ 0 & 1 & 6 & 0 \end{array} \right] R_2 \leftrightarrow R_3$$

$$\left[\begin{array}{ccc|c} 1 & 10 & -5 & 0 \\ 0 & 1 & 6 & 0 \\ 0 & -38 & 20 & 0 \end{array} \right] R_3 \rightarrow R_3 + 38R_2 \quad \left[\begin{array}{ccc|c} 1 & 10 & -5 & 0 \\ 0 & 1 & 6 & 0 \\ 0 & 0 & 248 & 0 \end{array} \right] R_3 \rightarrow \frac{R_3}{248}$$

$$\left[\begin{array}{ccc|c} 1 & 10 & -5 & 0 \\ 0 & 1 & 6 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right] R_1 \rightarrow R_1 + 5R_3 \quad \left[\begin{array}{ccc|c} 1 & 10 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right] R_1 \rightarrow R_1 - 10R_2$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right],$$

which is in reduced row Echelon form. The system of linear equations corresponding to above RREF is $\alpha_1 = 0 = \alpha_2 = \alpha_3$. Thus, we are getting only trivial solution and hence given set of vectors is linearly independent.

Exercise 15:

For Problems (a) – (i), determine whether the given set of vectors is linearly independent or linearly dependent in $\mathbb{R}^n$. In the case of linear dependence, find a dependency relationship.

$$(a) \left\{ \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\}$$

$$(b) \left\{ \begin{bmatrix} 2 \\ -1 \end{bmatrix}, \begin{bmatrix} 3 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$$

$$(c) \left\{ \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}$$

$$(d) \left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ -4 \\ 1 \end{bmatrix} \right\}$$

$$(e) \left\{ \begin{bmatrix} -2 \\ 4 \\ -6 \end{bmatrix}, \begin{bmatrix} 3 \\ -6 \\ 9 \end{bmatrix} \right\}$$

$$(f) \left\{ \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} \right\}$$

$$(g) \left\{ \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ -1 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} \right\}$$

$$(h) \left\{ \begin{bmatrix} 1 \\ -1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \right\}$$

$$(i) \left\{ \begin{bmatrix} 2 \\ -1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -1 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 3 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 2 \\ 1 \end{bmatrix} \right\}$$

Exercise 16:

Let $v_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, $v_2 = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$, $v_3 = \begin{bmatrix} 7 \\ 8 \\ 9 \end{bmatrix}$. Determine whether $\{v_1, v_2, v_3\}$ is linearly independent in $\mathbb{R}^3$.

Exercise 17:

Consider the vectors $v_1 = \begin{bmatrix} 2 \\ -1 \\ 5 \end{bmatrix}$, $v_2 = \begin{bmatrix} 1 \\ 3 \\ -4 \end{bmatrix}$, $v_3 = \begin{bmatrix} -3 \\ -9 \\ 12 \end{bmatrix}$ in $\mathbb{R}^3$. Show that $\{v_1, v_2, v_3\}$ is linearly dependent.

Exercise 18:

Determine all values of the constant k for which the vectors $\begin{bmatrix} 1 \\ 1 \\ k \end{bmatrix}$, $\begin{bmatrix} 0 \\ 2 \\ k \end{bmatrix}$ and $\begin{bmatrix} 1 \\ k \\ 6 \end{bmatrix}$ are linearly dependent in $\mathbb{R}^3$.

Exercise 19:

Test whether the vectors $p_1(x) = x^2 - 2x + 3$, $p_2(x) = 2x^2 + x + 8$, $p_3(x) = x^2 + 8x + 7$ are linearly independent.

Definition(Basis)

A subset B of a vector space V is said to be a basis for V if

- ✿ B is a spanning set of V , i.e., $\text{span}(B) = V$
- ✿ B is linearly independent subset of B .

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Example

$\left\{ e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \dots, e_n = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix} \right\}$ is a basis for $\mathbb{R}^n$. This basis is called the
standard basis or usual basis for $\mathbb{R}^n$.

Example

$\{1, x, x^2, \dots, x^n\}$ is a basis for $\mathbb{R}_n[x]$, known as the standard basis or usual basis for $\mathbb{R}_n[x]$.

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Example

$\varepsilon = \{E_{11}, E_{12}, E_{21}, E_{22}\}$ is a basis for $M_2(\mathbb{R})$, where

$$E_{11} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, E_{12} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, E_{21} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, E_{22} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

Example

Show that $B = \{1 + x, x + x^2, 1 + x^2\}$ is a basis for $\mathbb{R}_2[x]$.

Example

Show that $B = \{1 + x, x + x^2, 1 + x^2\}$ is a basis for $\mathbb{R}_2[x]$.

Solution

Let $a, b, c \in \mathbb{R}$ be such that

$$a(1 + x) + b(x + x^2) + c(1 + x^2) = 0 \implies (a + c) + (a + b)x + (b + c)x^2 = 0$$

Equating respective terms, we get the following set of equations

$$a + c = 0$$

$$a + b = 0$$

$$b + c = 0$$

Solving these equations, we get $a = b = c = 0$, which shows that B is a linearly independent set. For every $p(x) = a_0 + a_1x + a_2x^2 \in \mathbb{R}_2[x]$, we have

$$p(x) = \left[\frac{a_0 + a_1 - a_2}{2} \right] (1+x) + \left[\frac{-a_0 + a_1 + a_2}{2} \right] (x+x^2) + \left[\frac{a_0 - a_1 + a_2}{2} \right] (1+x^2).$$

Thus, $\text{span}(B) = \mathbb{R}_2[x]$ and hence B is a basis of $\mathbb{R}_2[x]$.

Exercise 20:

Show that $\left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right\}$ is a basis for $\mathbb{R}^3$.

Exercise 21:

Let $\mathbb{R}_2[x]$ be the vector space of polynomials of degree 2 or less. Prove that the set $\{1, 1+x, (1+x)^2\}$ is a basis for $\mathbb{R}_2[x]$.

(i) **Linearly independent:** Consider

$$\alpha \cdot 1 + \beta(1+x) + \gamma(1+x)^2 = 0 \implies (\alpha + \beta + \gamma) + (\beta + 2\gamma)x + \gamma x^2 = 0.$$

Comparing the coefficients of respective terms, we have

$$\gamma = 0 \quad \text{and} \quad \beta + 2\gamma = 0 \implies \beta = 0 \quad \text{and} \quad \alpha + \beta + \gamma = 0 \implies \alpha = 0$$

This shows that $\{1, 1+x, (1+x)^2\}$ is a linearly independent.

(ii) **Generator or Spanning set:** Let $p(x) = a_0 + a_1x + a_2x^2 \in \mathbb{R}_2[x]$. Consider

$$p(x) = a \cdot 1 + b(1+x) + c(1+x)^2 \implies p(x) = (a+b+c) + (b+2c)x + cx^2$$

Comparing the coefficients of respective terms, yields

$$\begin{aligned} c &= a_2 \\ b + 2c &= a_1 \implies b = a_1 - 2a_2 \\ a + b + c &= a_0 \implies a = a_0 - a_1 + a_2 \end{aligned}$$

Thus, $p(x) = a_0 - a_1 + a_2 + (a_1 - 2a_2)(1+x) + a_2(1+x)^2$. This shows that $\{1, 1+x, (1+x)^2\}$ is a generator of $\mathbb{R}_2[x]$.

Exercise 22:

Let

$$V = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \text{ in } \mathbb{R}^3 \mid x + 3y + z = 0 \right\} \quad \mathcal{B} = \left\{ \begin{pmatrix} -3 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ -3 \end{pmatrix} \right\}.$$

Verify that V is a subspace, and show directly that $\mathcal{B}$ is a basis for V .

Solution

First we observe that V is the solution set of the homogeneous equation $x+3y+z=0$, so it is a subspace. To show that $\mathcal{B}$ is a basis, we really need to verify three things:

1. **Both vectors are in V** because

$$\begin{array}{ccccccc} (-3) & + & 3(1) & + & (0) & = & 0 \\ (0) & + & 3(1) & + & (-3) & = & 0. \end{array}$$

2. **Span:** suppose that $\begin{pmatrix} x \\ y \\ z \end{pmatrix}$ is in V . Since $x+3y+z=0$, we have

$$y = -\frac{1}{3}(x+z),$$

so

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x \\ -\frac{1}{3}(x+z) \\ z \end{pmatrix} = -\frac{x}{3} \begin{pmatrix} -3 \\ 1 \\ 0 \end{pmatrix} - \frac{z}{3} \begin{pmatrix} 0 \\ 1 \\ -3 \end{pmatrix}.$$

Hence $\mathcal{B}$ spans V .

Solution cont. . .

3. **Linearly independent:**

$$c_1 \begin{pmatrix} -3 \\ 1 \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} 0 \\ 1 \\ -3 \end{pmatrix} = 0 \implies \begin{pmatrix} -3c_1 \\ c_1 + c_2 \\ -3c_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$
$$\implies c_1 = c_2 = 0.$$

Hence $\mathcal{B}$ is linearly independent.

Exercise 23:

Let V be the subspace defined by

$$V = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \mid x + 2y = z \right\}.$$

Find a basis for V . What is $\dim(V)$?

Definition(Coordinate)

Let $B = \{v_1, v_2, \dots, v_n\}$ be an **ordered** basis for a vector space $V(\mathbb{F})$ and let $v \in V$. Let

$$v = c_1 v_1 + c_2 v_2 + \cdots + c_n v_n.$$

Then the scalars $c_1, c_2, \dots, c_n$ are called the **coordinates of v** with respect to B , and the column vector

$$[v]_B = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}$$

is called the **coordinate vector** of v with respect to B .

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$$[v]_B = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}$$

is called the **coordinate vector** of v with respect to B .

Remark

Coordinate of a vector is always associated with an **ordered** basis.

Example

Find the coordinate vector of $p(x) = 1 - 3x + 4x^2$ with respect to ordered basis B of $\mathbb{R}_2[x]$, where

① $B = \{1, x, x^2\}$

② $B = \{1 + x, x + x^2, 1 + x^2\}$

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① $B = \{1, x, x^2\}$

② $B = \{1 + x, x + x^2, 1 + x^2\}$

Solution

① Since $p(x) = 1 - 3x + 4x^2$ and hence

$$[p(x)]_B = \begin{bmatrix} 1 \\ -3 \\ 4 \end{bmatrix}$$

② Since $p(x) = 1 - 3x + 4x^2 = -3(1 + x) + 4(1 + x^2)$ and hence

$$[p(x)]_B = \begin{bmatrix} -3 \\ 0 \\ 4 \end{bmatrix}$$

Result

Let $B = \{v_1, v_2, \dots, v_n\}$ be an ordered basis for a vector space V , let $u, v \in V$ and let $c \in \mathbb{F}$. Then

$$[u + v]_B = [u]_B + [v]_B \quad \text{and} \quad [cu]_B = c[u]_B$$

Result

Let $B = \{v_1, v_2, \dots, v_n\}$ be a basis for a vector space V . Then

- ① Any set of **more than n vectors** in V must be **linearly dependent**.
- ② Any set of **fewer than n vectors** in V can not be a **spanning set** of V .

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Result (The Basis Theorem)

If a vector space V has a basis with n vectors, then **every basis** for V has **exactly n vectors**.

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Result (The Basis Theorem)

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Definition(Dimension)

Let V be a vector space.

- ✿ The **dimension** of V , denoted **$\dim V$** , is the number of vectors in a basis for V . We write $\dim V = \infty$ if V does not have a finite basis.
- ✿ The dimension of the zero space **$\{0\}$** is defined to be **0**.

Remark

- ✿ The vector space V is called **finite-dimensional** if $\dim V$ is a **finite number**.
- ✿ The space V is called **infinite-dimensional** if there is **no finite** basis for V .

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Example

$\dim(\mathbb{R}^n) = n$, $\dim \mathbb{C}(\mathbb{C}) = 1$, $\dim \mathbb{C}(\mathbb{R}) = 2$, $\dim M_2(\mathbb{R}) = 4$ and $\dim \mathbb{R}_n[x] = n+1$.

Result

Let V be a vector space with $\dim V = n$.

- ① Any linearly independent set in V contains at most n vectors.
- ② Any spanning set for V contains at least n vectors.
- ③ Any linearly independent set of exactly n vectors in V is a basis for V .
- ④ Any spanning set for V of exactly n vectors is a basis for V .
- ⑤ Any linearly independent set in V can be extended to a basis for V .
- ⑥ Any spanning set for V can be reduced to a basis for V .

Example

Find a basis of the subspace

$$V = \text{Span} \left\{ \begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ -3 \\ 4 \end{pmatrix}, \begin{pmatrix} 0 \\ 4 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 5 \\ -2 \end{pmatrix} \right\}.$$

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Solution

Consider the following matrix A and converting it into row Echelon form, we get

$$A = \begin{bmatrix} 1 & -2 & 2 \\ 2 & -3 & 4 \\ 0 & 4 & 0 \\ -1 & 5 & -2 \end{bmatrix} \begin{array}{l} R_2 \rightarrow R_2 - 2R_1 \\ R_4 \rightarrow R_4 + R_1 \end{array} \begin{bmatrix} 1 & -2 & 2 \\ 0 & 1 & 0 \\ 0 & 4 & 0 \\ 0 & 3 & 0 \end{bmatrix} \begin{array}{l} R_3 \rightarrow R_3 - R_2 \\ R_4 \rightarrow R_4 - R_2 \end{array} \begin{bmatrix} 1 & -2 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix},$$

which is in row Echelon form and the first two columns are pivot columns, so a basis for V is

$$\left\{ \begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right\}.$$

Example

Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4, \mathbf{v}_5\}$ where

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 2 \\ -1 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 1 \\ 3 \\ 1 \\ 1 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} 1 \\ 5 \\ -1 \\ 5 \end{bmatrix}, \mathbf{v}_4 = \begin{bmatrix} 1 \\ 1 \\ 4 \\ -1 \end{bmatrix}, \mathbf{v}_5 = \begin{bmatrix} 2 \\ 7 \\ 0 \\ 2 \end{bmatrix}.$$

Let W be the subspace of $\mathbb{R}^4$. Find a basis for the subspace W and extend this basis to a basis of $\mathbb{R}^4$.

Solution

Let A be the matrix whose column vectors are vectors in the set S :

$$A = \begin{bmatrix} 1 & 2 & 2 & -1 \\ 1 & 3 & 1 & 1 \\ 1 & 5 & -1 & 5 \\ 1 & 1 & 4 & -1 \\ 2 & 7 & 0 & 2 \end{bmatrix}$$

Applying the elementary row operations to A to obtain its row Echelon form:

$$\begin{aligned} & \begin{bmatrix} 1 & 2 & 2 & -1 \\ 1 & 3 & 1 & 1 \\ 1 & 5 & -1 & 5 \\ 1 & 1 & 4 & -1 \\ 2 & 7 & 0 & 2 \end{bmatrix} \begin{array}{l} R_2 \rightarrow R_2 - R_1 \\ R_3 \rightarrow R_3 - R_1 \\ R_4 \rightarrow R_4 - R_1 \\ R_5 \rightarrow R_5 - 2R_1 \end{array} \begin{bmatrix} 1 & 2 & 2 & -1 \\ 0 & 1 & -1 & 2 \\ 0 & 3 & -3 & 6 \\ 0 & -1 & 2 & 0 \\ 0 & 3 & -4 & 4 \end{bmatrix} \begin{array}{l} R_3 \rightarrow R_3 - 3R_2 \\ R_4 \rightarrow R_4 + R_2 \\ R_5 \rightarrow R_5 - 3R_2 \end{array} \\ & \begin{bmatrix} 1 & 2 & 2 & -1 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & -1 & -2 \end{bmatrix} R_3 \leftrightarrow R_5 \quad \begin{bmatrix} 1 & 2 & 2 & -1 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & -1 & -2 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} R_4 \rightarrow R_4 + R_3 \quad \begin{bmatrix} 1 & 2 & 2 & -1 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & -1 & -2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{aligned}$$

Thus, a basis for W is

$$\left\{ \begin{bmatrix} 1 \\ 2 \\ 2 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -1 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ -1 \\ -2 \end{bmatrix} \right\}$$

and a basis for subspace $\mathbb{R}^4$ is

$$\left\{ \begin{bmatrix} 1 \\ 2 \\ 2 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -1 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ -1 \\ -2 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

Exercise 24:

Find a basis for $\text{Span}(S)$ where

$$S = \left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ -2 \\ -1 \end{bmatrix}, \begin{bmatrix} 2 \\ 6 \\ -2 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix} \right\}.$$

Exercise 25:

In each of the following, determine the dimension of the subspace of $\mathbb{R}^3$ spanned by the given vectors.

$$(a) \begin{bmatrix} 1 \\ -2 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ -2 \\ 4 \end{bmatrix}, \begin{bmatrix} -3 \\ 3 \\ 6 \end{bmatrix}$$

$$(b) \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}$$

$$(c) \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}, \begin{bmatrix} -2 \\ 2 \\ -4 \end{bmatrix}, \begin{bmatrix} 3 \\ -2 \\ 5 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}$$

Exercise 26:

Let W be the subspace of $\mathbb{R}^4$ spanned by the vectors

$$x_1 = \begin{bmatrix} 1 \\ -2 \\ 5 \\ -3 \end{bmatrix}, x_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 4 \end{bmatrix}, x_3 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

Find a basis for W and extend it to a basis for $\mathbb{R}^4$.

Exercise 27:

Let $\mathbb{R}_3[x]$ denote the set of polynomials of degree 3 or less with real coefficients. Consider the ordered basis $\mathcal{B} = \{1+x, 1+x^2, x-x^2+2x^3, 1-x-x^2\}$. Write the coordinate vector for the polynomial $f(x) = -3+2x^3$ in terms of the basis $\mathcal{B}$.

Exercise 28:

Determine whether each of the following sets is a basis for $\mathbb{R}^3$.

$$(a) S = \left\{ \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} -2 \\ 1 \\ 4 \end{bmatrix} \right\}.$$

$$(b) S = \left\{ \begin{bmatrix} 1 \\ 4 \\ 7 \end{bmatrix}, \begin{bmatrix} 2 \\ 5 \\ 8 \end{bmatrix}, \begin{bmatrix} 3 \\ 6 \\ 9 \end{bmatrix} \right\}.$$

$$(c) S = \left\{ \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 7 \end{bmatrix} \right\}$$

$$(d) S = \left\{ \begin{bmatrix} 1 \\ 2 \\ 5 \end{bmatrix}, \begin{bmatrix} 7 \\ 4 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 8 \\ 6 \end{bmatrix}, \begin{bmatrix} -1 \\ 9 \\ 10 \end{bmatrix} \right\}.$$